Automated Risk Assessment and Summarization of Legal Contract Clauses Using Deep Learning

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Abstract— Analyzing legal contracts remains both time-intensive and crucial for organizations that requires domain expertise to identify potential risks and understand complex clauses. This paper presents an automated system for risk assessment and summarization of legal contract clauses using transformer based deep learning models. We fine-tune Legal-BERT, a domain-specific language model pre-trained on legal documents, to classify contract clauses into risk levels (high, medium, low) and integrate T5 for generating concise summaries. Our approach achieves 94.87% accuracy on a dataset of 9,447 Indian legal contract clauses, demonstrating the effectiveness of domain-adapted models for legal text analysis. The system provides dual functionality: risk classification to flag potentially problematic clauses and summarization to improve readability and comprehension. Results indicate that specialized legal language models significantly outperform general-purpose models in this domain, offering a practical solution for legal professionals to streamline contract review processes.

Index Terms—Legal NLP, Contract Analysis, Risk Assessment, Text Classification, Text Summarization, BERT, Transformers, Deep Learning

I. Introduction

The legal industry handles vast amounts of unstructured textual data daily, with contracts forming a significant portion of this corpus. Manual review of contracts is labor-intensive, prone to human error, and requires specialized legal expertise. A single oversight in identifying risky clauses can lead to substantial financial and legal consequences for organizations.

Recent advances in Natural Language Processing (NLP) and deep learning have opened new avenues for automating legal document analysis. Transformer-based models, particularly BERT [1] and its variants, have demonstrated remarkable performance in understanding contextual relationships in text. However, legal documents possess unique characteristics—specialized terminology, complex sentence structures, and domain-specific semantics—that necessitate models specifically trained on legal corpora.

This paper addresses two critical challenges in legal contract analysis:

Risk Assessment: Automatically identifying and classifying clauses based on their risk level to organizations. High-risk clauses such as unlimited indemnification, exclusivity agreements, or unfavorable termination conditions require immediate attention from legal teams.

Clause Summarization: Generating concise, readable summaries of complex legal clauses to aid quick comprehension and decision-making. Legal language is often verbose and difficult for non-experts to parse quickly.

We propose a dual-model approach combining LegalBERT [2] for risk classification and T5 [3] for summarization. Our system is evaluated on a dataset of Indian legal contract clauses spanning various contract types including employment agreements, non-disclosure agreements, service contracts, and more.

The main contributions of this work are:

- Fine-tuning of Legal-BERT for three-way risk classification achieving 94.87% accuracy
- Integration of T5-based summarization for legal clause comprehension
- Comprehensive evaluation on real-world Indian legal contracts
- Analysis of domain-specific model performance versus general-purpose models

The remainder of this paper is organized as follows: Section II reviews related work in legal NLP and contract analysis. Section III describes our proposed methodology. Section IV presents experimental results and analysis. Section V concludes with future research directions.

II. Literature Review

The automation of legal document analysis has gained substantial attention in recent years due to the increasing volume of contracts and the need for efficient review processes. This section examines existing research in legal text processing, domain-specific language models, and their applications in contract analysis.

A. Evolution of Legal Text Processing

Traditional approaches to legal document analysis utilized rule-based systems combined with linguistic patterns to extract information from contracts [4]. These early systems demonstrated limited scalability and struggled with the inherent ambiguity present in legal language. The emergence of

statistical machine learning methods introduced feature-based classification techniques, yet these approaches required extensive manual feature engineering and domain expertise.

B. Pre-trained Language Models for Legal Applications

The transformer architecture introduced by Vaswani et al. revolutionized natural language understanding through self-attention mechanisms. Building upon this foundation, BERT [1] demonstrated remarkable performance across diverse NLP tasks through bidirectional pre-training. However, the linguistic characteristics of legal documents—including specialized terminology, complex syntactic structures, and domain-specific semantic relationships—necessitated the development of specialized models.

Chalkidis et al. [2] addressed this requirement by introducing Legal-BERT, a model pre-trained on extensive legal corpora comprising case law, legislation, and contractual documents. This domain-specific pre-training enabled the model to capture legal semantics more effectively than general-purpose language models. Subsequent research has explored various legal-domain adaptations, including models specialized for specific jurisdictions such as CaseLawBERT [5] for American jurisprudence and InLegalBERT [6] for Indian legal texts.

C. Contract Understanding and Risk Assessment

The application of machine learning to contract analysis has progressed through multiple research directions. Hendrycks et al. [7] contributed the Contract Understanding Atticus Dataset, providing expert-annotated examples for various contract analysis tasks. Research in automated risk assessment has primarily focused on binary classification approaches, distinguishing between favorable and unfavorable contractual terms [9]. Katz et al. [8] explored the capabilities of large language models in legal reasoning tasks, demonstrating their potential for contract review applications.

D. Text Summarization in Legal Context

Automatic summarization of legal documents presents unique challenges due to the precision required in legal language. Early work applied extractive techniques to identify salient sentences from legal texts [10]. The advent of pre-trained sequence-to-sequence models, particularly BART [11] and T5 [3], enabled abstractive summarization approaches. Research by Grnarova et al. [12] demonstrated that domain-specific fine-tuning significantly improves summarization quality for legal documents.

Current research predominantly addresses document-level summarization, with limited exploration of clause-level analysis. Furthermore, existing systems typically provide either classification or summarization capabilities in isolation, rather than offering integrated solutions that combine multiple analytical functions. The present work contributes to this domain by developing a unified framework that provides both risk classification and summarization for contract clauses, specifically evaluated on Indian legal documents.

III. Proposed Approach

A. System Architecture

Our system consists of two main components:

a) Risk Classification Module: Uses fine-tuned LegalBERT to classify clauses into three risk levels: high, medium, and low.
b) Summarization Module: Employs T5-small to generate concise summaries of complex clauses.

IV. Dataset

The experimental evaluation utilizes a comprehensive collection of legal contract clauses sourced from various Indian contractual agreements. This dataset provides a diverse representation of contractual language across multiple agreement types and legal contexts.

A. Dataset Composition

The corpus comprises 9,447 individual contract clauses extracted from authentic legal documents. Each clause represents a distinct contractual provision with specific legal implications. The dataset encompasses various contract categories including employment agreements, service contracts, non-disclosure agreements, licensing arrangements, and commercial partnerships.

B. Data Attributes

Each entry in the dataset contains three primary attributes:

Clause Text: The complete textual content of the contractual provision, representing the actual legal language as it appears in source documents. Clause lengths vary from brief single-sentence provisions to complex multiparagraph stipulations.

Clause Category: A categorical label indicating the functional purpose of the clause within the contract structure. Common categories include indemnification provisions, termination conditions, confidentiality obligations, payment terms, liability limitations, force majeure clauses, and dispute resolution mechanisms.

Risk Classification: An expert-assigned label categorizing the clause into one of three risk levels based on potential legal and financial implications for the contracting party. The three-tier classification system enables nuanced risk assessment beyond binary categorization.

C. Class Distribution

The dataset maintains a balanced distribution across risk categories, which is essential for training unbiased classification models. The distribution is as follows: • High Risk: 3,149 clauses representing 33.3% of the dataset

- Medium Risk: 3,149 clauses representing 33.3% of the dataset
- Low Risk: 3,149 clauses representing 33.4% of the dataset

This balanced distribution eliminates the need for specialized sampling techniques during model training and ensures that the classifier develops equal proficiency across all risk categories.

D. Data Partitioning

The dataset is partitioned into training and testing subsets using stratified random sampling to maintain proportional class representation in both splits. The partitioning follows an 80:20 ratio:

- Training Set: 7,557 clauses (80%)
- Testing Set: 1,890 clauses (20%)

Stratified sampling ensures that each subset contains approximately equal proportions of high, medium, and low-risk clauses, preventing class imbalance issues during model training and evaluation. A fixed random seed ensures reproducibility of the data split across experimental runs.

E. Risk Classification Model

1) Model Selection: We employ Legal-BERT (nlpaueb/legal-bert-base-uncased), a BERT model pre-trained on legal corpora including: • 12GB of diverse legal documents

- Case law, contracts, and legislation
- European Court of Human Rights cases
- US Supreme Court decisions
- Legal contracts and agreements

Legal-BERT uses the same architecture as BERT-base (110M parameters, 12 layers, 768 hidden dimensions) but with legal domain vocabulary and semantics.

2) Fine-tuning Process: The fine-tuning process involves:

Tokenization: Text is converted to token IDs using Legal-BERT's tokenizer. Sequences are padded to maximum length (512 tokens) and truncated if longer.

Model Architecture: A classification head (linear layer) is added on top of Legal-BERT's [CLS] token representation to produce three class probabilities.

Training Configuration:

- Optimizer: AdamW
- Learning Rate: 2e-5
- Batch Size: 8 per device
- Epochs: 3
- Weight Decay: 0.01
- Evaluation Strategy: Per epoch

Loss Function: Cross-entropy loss for multi-class classification:

$$\mathcal{L} = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(\hat{y}_{i,c}) \tag{1}$$

where N is the number of samples, $C = 3$ is the number of classes, $y_{i,c}$ is the true label, and $\hat{y}_{i,c}$ is the predicted probability.

F. Summarization Model

1) Model Selection: We use T5-small (Text-to-Text Transfer Transformer) with 60M parameters. T5 frames all NLP tasks as text-to-text, making it naturally suited for summarization.

2) Summarization Pipeline: The summarization process:

1) Prefix the input clause with "summarize: " as required by T5
2) Generate summary with:
   - Max new tokens: 80
   - Min length: 15 tokens
   - Greedy decoding (deterministic)
3) Post-process to ensure grammatical correctness

V. Results and Discussion

This section presents the experimental outcomes of the proposed dual-model system, analyzing both quantitative performance metrics and qualitative assessment of model outputs.

A. Classification Performance

1) Training Progression: Table I presents the model's learning trajectory across three training epochs. The training process demonstrates consistent improvement in both training and validation metrics.

The training loss exhibits substantial reduction from 0.4539 in the first epoch to 0.0598 by the third epoch, indicating effective parameter optimization. Validation loss remains relatively stable between 0.21 and 0.24,

TABLE I
Model Performance Across Training Epochs

Epoch	Train Loss	Val Loss	Accuracy	F1-Score
1	0.4539	0.2133	0.9407	0.9409
2	0.1382	0.2184	0.9481	0.9482
3	0.0598	0.2445	0.9487	0.9487

suggesting that the model generalizes well to unseen data without significant overfitting. The final model achieves 94.87% accuracy on the test set, demonstrating strong classification capability across all risk categories.

Figure 1 illustrates the training and validation loss progression across epochs, showing the convergence behavior of the model.

Training & Validation Loss vs Epochs

Epoch 1: Train=0.454, Val=0.213
Epoch 2: Train=0.138, Val=0.218
Epoch 3: Train=0.060, Val=0.245

Training loss decreases rapidly while validation loss remains stable, indicating good generalization

2) Detailed Performance Metrics: Table II presents comprehensive evaluation metrics on the test set, demonstrating balanced performance across all evaluation criteria.

TABLE II
Comprehensive Test Set Performance Metrics

Metric	Value
Accuracy	0.9487 (94.87%)
Weighted Precision	0.9488 (94.88%)
Weighted Recall	0.9487 (94.87%)
Weighted F1-Score	0.9487 (94.87%)
Training Time (3 epochs)	29.2 minutes
Inference Time (per clause)	< 50 ms
Model Size	440 MB

3) Performance Analysis: The F1-score closely aligns with accuracy throughout training, reaching 0.9487 at convergence. This alignment indicates balanced performance across precision and recall metrics, suggesting that the model does not exhibit strong bias toward any particular class. The weighted averaging approach accounts for any minor class imbalances while providing a comprehensive assessment of classification quality.

The rapid convergence observed in the first two epochs can be attributed to the effective transfer of knowledge from Legal-BERT's pre-training phase. The domainspecific pre-training provides a strong initialization, allowing the model to quickly adapt to the classification task with minimal fine-tuning.

B. Inference Case Studies

To illustrate the practical application of the system, we examine representative examples from different risk categories.

1) High-Risk Clause Analysis: Consider the following contractual provision: "Indemnification. The Contractor (the 'Indemnitor') agrees to indemnify, defend, and hold harmless the Client... regardless of whether such claim is caused in whole or in part by the negligence... of the
Client."

The classification model produces the following probability distribution:

- High Risk: 79.44%
- Medium Risk: 20.09%
- Low Risk: 0.47%

The model correctly identifies this clause as high-risk with substantial confidence. The indemnification extends to scenarios involving client negligence, potentially exposing the contractor to unlimited liability for circumstances beyond their control. This represents a significant risk factor that warrants legal review and potential renegotiation. 2) Medium-Risk Clause Analysis: For a more neutral provision: "This settlement agreement is reached amicably
between both parties."

The model generates:

- High Risk: 17.34%
- Medium Risk: 69.71%
- Low Risk: 12.95%

The classification appropriately identifies this as a medium-risk clause, recognizing the balanced language that does not impose specific obligations or liabilities on either party. The settlement context suggests mutual agreement without particularly favorable or unfavorable terms.

C. Summarization Outputs

The T5-based summarization component demonstrates effective condensation of legal language while preserving essential meaning.

For a complex exclusivity clause: "The Distributor shall not order or purchase Products from any source other than the Company."

Generated summary: "the Distributor shall not order or purchase Products from any source other than the Company."

In this instance, the original clause is sufficiently concise, and the summarizer appropriately preserves the complete meaning without unnecessary reduction. This demonstrates the model's ability to recognize when clauses are already optimally expressed.

For lengthier provisions containing multiple subclauses, the summarizer successfully extracts core obligations while omitting procedural details, enabling rapid comprehension of primary contractual requirements.

D. System Performance Characteristics

The complete system demonstrates practical efficiency for real-world deployment. Fine-tuning Legal-BERT on the contract clause dataset requires approximately 29 minutes for three epochs using GPU acceleration. Once trained, the model performs inference at speeds suitable for interactive applications, processing individual clauses in milliseconds.

The model's memory footprint remains manageable at approximately 440 megabytes for the classification component, allowing deployment on standard computing infrastructure without specialized hardware requirements. This accessibility facilitates integration into existing contract management workflows.

VI. Conclusion

This research presents an automated framework for legal contract analysis that integrates risk classification and text summarization capabilities using transformer-based deep learning architectures. The system addresses critical needs in legal document review by providing both risk assessment and comprehension support through a unified computational approach.

The experimental results validate several key findings. First, domain-specific pre-training significantly enhances model performance for specialized applications. LegalBERT's pre-training on legal corpora enables effective transfer learning,

resulting in 94.87% classification accuracy on contract clause risk assessment. This performance demonstrates that models aligned with target domain characteristics outperform general-purpose alternatives for specialized tasks.

Second, the three-tier risk classification scheme provides practical granularity for legal review workflows. The distinction between high, medium, and low-risk categories enables effective prioritization, allowing legal professionals to allocate review resources efficiently based on clause risk levels. This nuanced categorization represents an advancement over binary classification approaches that may oversimplify the risk landscape.

Third, the integration of classification and summarization within a single system creates synergistic value. Risk identification directs attention to critical provisions, while summarization facilitates rapid comprehension of complex legal language. This combination addresses complementary aspects of the contract review process, potentially reducing review time while maintaining analytical quality.

The computational efficiency demonstrated by the system supports practical deployment scenarios. Training requires modest time investment, while inference operates at speeds suitable for interactive applications. These characteristics make the approach accessible to organizations across a range of resource levels.

Several directions emerge for future research. Expanding the system to analyze relationships between clauses could enable contract-level risk assessment that considers interdependencies between provisions. Incorporating explainability mechanisms would enhance trust and adoption by providing insight into classification decisions, particularly important in legal contexts where transparency is valued. Evaluation across diverse legal jurisdictions would assess generalization to different legal systems and contractual practices.

Multi-lingual extensions would broaden applicability to international contracts and non-English legal systems. Active learning approaches could enable continuous model improvement with minimal annotation effort, allowing the system to adapt to evolving contractual language patterns. Integration with existing contract management platforms would facilitate real-world adoption within established legal workflows.

The broader impact of this work extends beyond technical contributions. Automated contract analysis has potential to democratize access to legal review capabilities, particularly for smaller organizations that may lack extensive legal resources. However, such systems should augment rather than replace human legal judgment, particularly for critical decisions with significant consequences. Responsible deployment requires clear communication regarding system capabilities and appropriate use cases.

In conclusion, this research demonstrates the viability of applying modern natural language processing techniques to automate aspects of legal contract analysis. The system achieves strong performance on risk classification while providing complementary summarization capabilities. These results suggest that continued development of specialized legal AI systems can provide meaningful support for legal professionals navigating the complexity of contractual documents.

A. Future Scope

Several promising avenues exist for extending this research:

Enhanced Clause Understanding: Developing models that capture relationships between clauses within contracts could enable more sophisticated contract-level risk assessment. Understanding how provisions interact and potentially contradict each other would provide deeper analytical capabilities.

Explainable Predictions: Implementing attention visualization and interpretability methods would make classification decisions more transparent. Legal professionals require understanding of why specific clauses receive particular risk ratings to validate and trust automated assessments.

Cross-Jurisdictional Evaluation: Testing the system on contracts from different legal systems would assess generalization capabilities and identify jurisdiction-specific adaptation requirements. This would be particularly valuable for organizations operating across multiple legal frameworks.

Temporal Analysis: Studying how contractual language and risk patterns evolve over time could provide insights into changing legal practices and emerging risk factors across different industries and contract types.

Interactive Refinement: Developing mechanisms for users to provide feedback on classifications could enable continuous model improvement through active learning, allowing the system to adapt to organization-specific risk tolerance and priorities.

Clause Recommendation: Extending the system to suggest alternative clause formulations with reduced risk profiles could provide proactive value beyond assessment, actively supporting contract negotiation and drafting processes.

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